SolarPro Global — Technical Guide

Engineering basis, standards, and integration reference

Version: 2026-06-05 (beta) **Audience:** Senior engineers, technical leads, and procurement/IT staff at Ghana installer and supplier companies; internal engineering team. **Companion to:** User Guide (workflow) and Sales Pitch (positioning).

> *This document explains **how** SolarPro computes its outputs and **which standards** it follows, so an external engineer can defend any value the platform produces to a client or to a financing bank.*

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1. Engineering principles in one page

SolarPro is built on three rules:

1. **Every output is derivable from the inputs.** No "magic" multipliers — every coefficient is listed in this guide or in `assumptions.md`.
2. **Every output is overridable.** Defaults are conservative; the engineer can replace any value with their own.
3. **Every output is traceable.** The proposal PDF shows the calculation path so a reviewing engineer can validate without re-running the tool.

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2. Standards applied

Standard	Where it's applied	Notes
BS 7671 (18th Edition, +A2)	AC cable sizing, derating, protective device coordination	UK origin; default standard for Ghana commercial/industrial wiring
IEC 62548	DC string design, MPPT voltage windows, isolation	International PV-array standard
IEC 60364-7-712	PV-specific installation requirements	Used for protective bonding, isolation
IEC 61730	Module safety classification	Determines Class II isolation requirements
NEC Article 690	Fallback for US-style installations	Triggered only when site country = USA
Ghana Energy Commission — Solar PV Code of Practice (2019)	Permitting checklist on the installation report	Drives the EC-certification section

IEC 61724	Performance ratio & yield calculation	Drives the economic analysis

Standards are cited inline in every report so the bank's engineer can verify.

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3. Site assessment

3.1 Irradiance

Source: `config/global_solar_data.py` — region-level peak sun hours (PSH) lookup.

For Ghana:

Region	PSH (kWh/m²/day)	Ambient (°C)
Greater Accra	5.2	28
Ashanti	5.0	26
Northern	5.6	30
Volta	5.1	27
Western	4.8	26

Source data: NASA POWER + Ghana Meteorological Agency monthly means, averaged 2018–2023.

3.2 ECG tariff

Region-tagged residential and commercial tariffs sourced from Public Utilities Regulatory Commission (PURC) quarterly publications. Escalation default = 6%/year (BoG inflation forecast).

The engineer can override tariff and escalation in the economic step.

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4. Load analysis

```
Daily energy demand (Wh/day) = Σ(P_i × n_i × h_i × d_i / 7)
```

Where:

- `P_i` = nameplate power of load `i` (W)
- `n_i` = quantity
- `h_i` = hours per day
- `d_i` = days per week (default 7)

A diversity factor is applied to the **peak** load (not the daily energy):

```
Peak (kW) = Σ(P_i × n_i) × DF
```

Where `DF` = 0.65 for residential, 0.85 for commercial, 1.0 for industrial. The engineer can change `DF` per project.

Surge load = peak × 3 (motor start; configurable 2–5×).

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5. PV array sizing

5.1 Required peak power

```
P_pv (kWp) = E_daily / (PSH × η_system × δ_temp)
```

Where:

- `E_daily` = daily energy demand (kWh)
- `PSH` = peak sun hours from §3.1
- `η_system` = 0.80 (default) — combined inverter + wiring + soiling
- `δ_temp` = $1 - \gamma \times (T\_cell - 25)$ — temperature derate

Cell temperature: `T_cell = T_amb + 25 °C` (NOCT approximation; configurable). γ (power temperature coefficient) = $-0.0034\ K^{-1}$ (Tier-1 mono panel default).

5.2 Module count

```
N_modules = ceil(P_pv × 1000 / W_module)
```

Where `W_module ∈` {110, 250, 330, 400, 450, 500, 550} Wp (commit `27b2492`).

5.3 String design

For each candidate inverter MPPT range `[V_min, V_max]`:

```
N_series_min = ceil(V_min / (V_mp × δ_T_hot))
N_series_max = floor(V_max / (V_oc × δ_T_cold))
```

Temperature corrections:

- `δ_T_hot = 1 + β_voc × (T_cell_hot − 25)` ≈ 0.88 (Vmp shrinks when panel is hot)
- `δ_T_cold = 1 + β_voc × (T_cold − 25)` ≈ 1.10 (Voc grows when panel is cold)

`β_voc` = $-0.0029\ K^{-1}$ default.

If `N_series_min > N_series_max`, the MPPT panel is incompatible → engine flags WARNING and suggests a smaller-Vmp module or a different inverter.

5.4 Mounting derate

Mounting type	Soiling factor	Temperature uplift (°C)
`rooftop_pitched`	0.95	+5
`rooftop_flat`	0.90	+10
`rooftop_metal`	0.95	+12
`rooftop_membrane`	0.95	+8
`ground_fixed`	0.92	+3
`ground_tracking`	0.95	+3

These factors fold into `η_system` and `T_cell` automatically.

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6. Battery sizing

```
C_battery (Wh) = E_daily × N_autonomy / (DoD × η_inv × η_batt)
```

Defaults:

- `N_autonomy` = 1 day (residential), 0.5 day (grid-tied), 2 days (off-grid critical)
- `DoD` = 0.80 (lithium), 0.50 (gel/AGM)
- `η_inv` = 0.95
- `η_batt` = 0.95 (lithium), 0.85 (lead-acid)

The battery type (lithium / gel / AGM) drives DoD and η_batt. The engineer chooses; the engine reflects the math.

Output is rounded up to the nearest stocked battery module size (5 kWh, 10 kWh, 15 kWh, 20 kWh by default — configurable in Settings → BOQ).

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7. Inverter sizing

```
P_inv (kW) = max(Peak_kW × 1.25, P_pv × 0.85)
```

Two-sided constraint:

- Must handle peak load with 25% margin
- Must absorb at least 85% of PV array output (Ghana standard avoids over-clipping under low PSH)

A hybrid inverter is selected by default for off-grid + battery; grid-tied selects a string inverter.

8. Cable sizing

8.1 DC cable (panel → combiner → inverter)

Voltage drop method (IEC 62548):

```
A (mm²) = (2 × L × I × ρ) / ΔV_max
```

Where:

- `L` = one-way length (m)
- `I` = string current at STC
- `ρ` = 0.0175 Ω·mm²/m (Cu @ 20 °C)
- `ΔV_max` = 1.5% × V_array

Round up to the next standard size (4, 6, 10, 16, 25 mm²). Verify the result against ampacity at 70 °C ambient — bump up one size if borderline.

8.2 AC cable (inverter → DB → load)

Per BS 7671 §523, Table 4D5 (PVC SWA):

Cable	Current carrying capacity (A) @ 30 °C ambient
2.5 mm²	24 A
4 mm²	32 A
6 mm²	41 A
10 mm²	57 A
16 mm²	76 A
25 mm²	101 A

Derating applied:

- Ambient temperature factor (BS 7671 Table 4B1) — Ghana ambient 32–35 °C → 0.94
- Grouping factor (Table 4C1) — if more than one cable shares conduit
- Installation method factor (clipped direct vs. conduit)

Final: smallest size whose derated capacity ≥ design current × 1.25.

Voltage drop check: ≤ 3% for final circuits, ≤ 5% combined (per BS 7671 Appendix 12).

8.3 Earthing

Earth conductor sized per BS 7671 §544 — half the phase conductor's cross-section, minimum 4 mm² Cu.

Earth rod: 1.2 m × 16 mm Cu — measured resistance ≤ 10 Ω per IEC 60364-7-712.

9. Bill of Quantities — how it's built

`calc_boq()` traverses the sized system and emits one line per material category:

Category	Quantity logic
Modules	`N_modules` from §5.2
Mounting rails	`ceil(N_modules × 1.05 m / 4.2 m)` — 4.2 m standard rail
End clamps	`4 × N_strings`
Mid clamps	`2 × (N_modules − N_strings)`
MC4 connectors	`2 × N_strings`
DC cable	length from §8.1 with 10% wastage
AC cable	length from §8.2 with 10% wastage
Inverter	`N_inv`
Battery modules	`ceil(C_battery / W_battery_unit)`
Earth rod	1 per system
Earth conductor	length matches AC cable
Surge arrestor (DC)	1 per MPPT input
Surge arrestor (AC)	1 per phase
Isolators (DC)	1 per string
Combiner box	1 per inverter
Ground mount only	steel post (1 per 4 modules), concrete footing (1 per post), purlin beam (4.2 m per row), earth rod (1 per 20 modules)

Pricing:

- Default: blank — engineer types unit cost.
- With supplier integration enabled: pulls live pricing via the supplier's API. Currently scaffolded for Nocheski and Ozo (target Q3 2026).

10. Economic analysis

10.1 Energy yield (per IEC 61724)

```
E_year (kWh) = P_pv × PSH × 365 × PR
```

`PR` (performance ratio) defaults:

- Rooftop residential: 0.75
- Rooftop commercial: 0.80
- Ground mount fixed: 0.82
- Ground mount tracking: 0.86

10.2 Cash flow

For year `y` ∈ [1, lifetime]:

```
Savings_y = E_year × (1 − δ_y) × Tariff × (1 + escalation)^(y − 1)
OPEX_y    = OPEX_0 × (1 + 0.03)^(y − 1)
Net_y     = Savings_y − OPEX_y
NPV_y     = Net_y / (1 + discount)^y
```

Where:

- `δ_y` = degradation per year (default 0.5%, lithium battery system; 0.7% pure PV)
- `OPEX_0` = annual maintenance estimate (default 1% of CAPEX)

10.3 Headline metrics

- **Payback (years)** — interpolated to the year where cumulative cashflow ≥ 0
- **IRR (%)** — Newton-Raphson on the cashflow vector, tolerance 1e-6
- **NPV (GHS)** — sum of `NPV_y` over lifetime
- **LCOE (GHS/kWh)** — `Σ Costs / Σ Energy`

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11. Reports

Every project produces seven PDF reports, all rendered from the same JSON state:

1. **PV Sizing Report** — array, strings, MPPT match, temperature checks
2. **BOQ** — full materials list with prices
3. **Cable Schedule** — every run with gauge, length, derating, voltage drop
4. **Economic Analysis** — charts + IRR/NPV table
5. **Installation Method Statement** — step-by-step, with safety
6. **Energy Yield Report** — month-by-month projection
7. **Bankable Proposal** — combined cover-page + executive summary + sized excerpts of 1–6

Reports embed the assumptions used (PSH, system efficiency, derating factors) so a third-party engineer can validate without re-running the tool.

12. Data model (for IT staff)

12.1 Storage

Each project's full engineering state lives in a single JSON blob: `projects.data_json`. Schema (abbreviated):

```
{
  "site": { "country": "Ghana", "region": "Greater Accra", "psh": 5.2, "ambient_c": 28,
  "loads": [ { "name": "Fridge", "watts": 250, "qty": 1, "hours": 12, "days_per_week":
  "totals": { "daily_kwh": 18.3, "peak_kw": 4.1, "surge_kw": 12.3 },
  "results": {
    "pv":      { "p_kwp": 6.0, "n_modules": 12, "w_module": 500, "strings": [...] },
    "battery":{ "kwh": 10, "type": "lithium" },
    "inverter":{ "kw": 5.0, "type": "hybrid" },
    "cable":  { "dc_mm2": 6, "ac_mm2": 4 },
    "boq":     [...],
    "economics":{ "payback_y": 4.2, "irr": 0.27, "npv_25y": 86_200 }
  }
}
```

12.2 Export

- **PDF** — `Generate Proposal` button on the project page.
- **JSON** — `GET /api/project/<id>/export.json` (authenticated; returns the schema above).
- **CSV** — `GET /api/project/<id>/boq.csv` (BOQ only).

12.3 Re-import

POST a previously exported JSON to `/api/project/import` — recreates the project. Useful for restoring from backup or moving a project between accounts.

13. Integration points

For supplier and installer companies, three integration surfaces are available during beta:

13.1 Pricing API (suppliers)

If you sell panels / inverters / batteries / cables, we'll wire your stock into the BOQ procurement screen. Required from you:

- A REST or GraphQL endpoint returning JSON with `{ sku, name, category, unit_price_ghs, in_stock, lead_time_days }`

- Optional webhook for stock updates

We pull on a 6-hour cadence (configurable). Live unit costs replace blank cells in the BOQ.

13.2 Project handoff (installers)

After a designer signs off a project, a `/installer/handoff` webhook fires to your endpoint with the full JSON state. Your project management system picks it up and creates a work order.

13.3 Single sign-on (SSO)

If your company uses Google Workspace, Microsoft 365, or any OIDC provider — we support SSO on Business+ plans. Required: client ID + client secret + issuer URL.

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14. Security & compliance

Control	Implementation
Authentication	Username + password; 5-failed-attempt lockout 15 min
Session management	HttpOnly Lax cookie; CSRF (`_csrf`) on every POST
Database isolation	Per-tenant row-level filters on all user data tables
Encryption at rest	Hosting platform-managed (currently AES-256)
Encryption in transit	TLS 1.2+ end-to-end
Audit log	`audit.log` (JSON-line) — login, project edits, exports
Security headers	CSP, X-Content-Type-Options, Referrer-Policy, Frame-Ancestors
Data export	Self-serve from Settings → Export (per GDPR Art. 20)
Data deletion	Self-serve from Settings → Delete Account
Backups	Daily snapshot, 30-day retention

Reference: `SECURITY.md` in the repo for the full Zero-Trust architecture + RBAC matrix.

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15. Performance & SLA targets (beta)

Metric	Target	Notes
Sizing calculation latency	< 30 s	p95
Proposal PDF generation	< 60 s	p95
Uptime	99.0%	Free Railway tier; will rise to 99.9% on K8s

Support response (in-app chat)	< 60 s	Rule-based; AI fallback to Claude/Llama
Support response (email)	< 24 h	Business hours

16. Where to look in the code (internal staff)

Concern	File
Loads → sizing pipeline	`web_app.py` → `calc_loads` / `calc_pv` / `calc_battery`
AC cable sizing	`calculation/ac_cable_sizing.py`
DC cable sizing	`web_app.py` → `size_dc_cable`
BOQ generation	`web_app.py` → `calc_boq`
Economic engine	`web_app.py` → `calc_economics`
Global irradiance data	`config/global_solar_data.py`
AI helpline	`web_app.py` → `/api/assistant/chat` + `api_manager.py`
Email send chain	`api_manager.py` → `_send_email`
Structured logging	`logging_config/structured_logger.py`
Health endpoints	`web_app.py` → `/api/health*`

17. Roadmap (next 90 days)

Month	Theme	Highlights
June 2026	Beta hardening	Custom domain SSL, multi-tenant edge isolation, screenshots in user guide
July 2026	Procurement	Nocheski + Ozo pricing live in BOQ, supplier directory page
August 2026	Energy reporting	Month-by-month yield, Excel export of BOQ, mobile-optimised dashboard

Feature requests during beta: send to `support@aiappinvent.com` or via in-app chat. Top-3 by demand each fortnight ships on the next sprint.

18. Screenshots index

The technical guide references the same screens as the user guide:

- `[SCREEN: location-globe]` — for §3 site assessment
- `[SCREEN: loads-input]` — for §4 load analysis

- `[SCREEN: results-pv]` — for §5 PV sizing
- `[SCREEN: boq]` — for §9 BOQ
- `[SCREEN: economics]` — for §10 economics
- `[SCREEN: installation-drawings]` — for §6.4 mounting derate consequences

Capture from the live app, save under `docs/screens/<slug>.png`, rerun the PDF build to embed.